

Technology Stack

LAYER	TECHNOLOGY	PURPOSE
Mobile App	React Native (Expo)	Cross-platform Android/iOS app
Backend API	ASP.NET Core 8	REST API + SignalR real-time hub
Database	SQL Server	Primary data store (on VPS)
Offline DB	SQLite (expo-sqlite)	Local cache for offline mode
Auth	JWT Tokens	Custom login (no Google OAuth)
Real-time	SignalR + Firebase (FCM)	Instant order alerts + push notifications
Location	expo-location	GPS tracking at visit & order
PDF	QuestPDF	GST invoice PDF generation

System Architecture

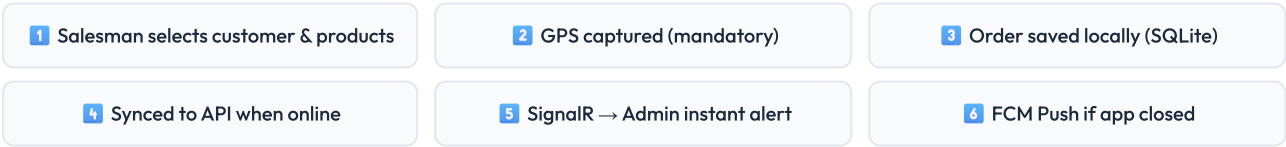
**Mobile App**
React Native → Local SQLite (offline) → Sync Engine → REST API

**ASP.NET API**
10 Controllers + SignalR Hub + GST Engine + PDF Generator. JWT on all endpoints.

**Sync Engine**
Queues offline ops, auto-pushes on reconnect. UUID dedup prevents duplicates.

**SQL Server**
11 tables: Users, Customers, Products, Orders, OrderItems, Invoices, InvoiceItems, LocationLogs, Notifications, States, SyncQueue

Data Flow: Order Placement



Confirmed Decisions

#	TOPIC	DECISION
1	Salesman Registration	Admin-only — Admin creates accounts, shares credentials
2	Customer Management	Salesmen CAN add new customers in the field
3	Offline Mode	Full offline support + auto-sync on reconnect
4	GST Billing	Full Indian GST — CGST+SGST (intra) / IGST (inter), per-product rates
5	Notifications	SignalR + FCM — dual system for max speed
6	Hosting	Own VPS
7	Scale	3 Admins + 10 Salesmen, architect for 100 concurrent